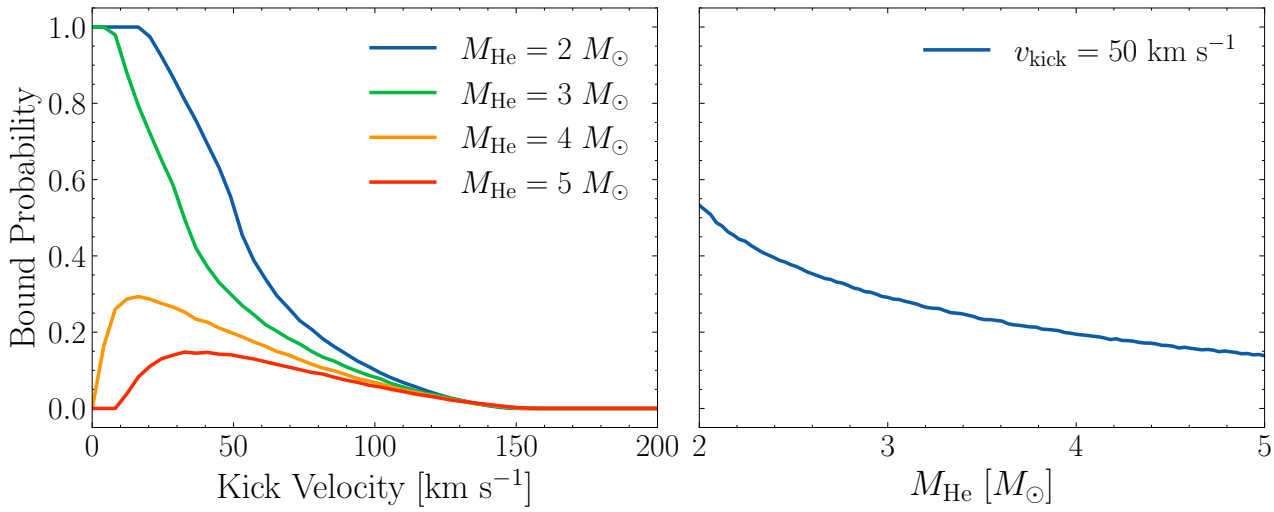

Figure 11. Probability that orbit stays bound as a function of natal kick and He star mass. We assume a default initial orbital period of 100 days, a red giant mass of 1 $M_\odot$, and a neutron star mass of 1.4 $M_\odot$. For massive He stars and/or large natal kicks, fine tuning is required to prevent the red giant + NS binary from becoming unbound.

4U 1700+24, which has an orbital period of 4391 days (Hinkle et al. 2019), lies at a distance of just ~ 0.4 kpc (Masetti et al. 2002). This suggests that NS + MS binaries with orbital periods too long to be detectable by *Gaia* may be even more common then the ones we consider here.

7. CONCLUSION

IGR J16194-2810 is a confirmed member of the rare class of symbiotic X-ray binaries, in which a compact object accretes from an evolved low-mass star. IGR J16194-2810 has been studied extensively in the X-rays and was suspected to host a neutron star based on its X-ray phenomenology. In this work, we present a dynamical mass measurement of the neutron star with optical spectroscopic follow-up. We synthesize our observations with archival photometry to determine the orbital period of the system, which is 192.73 days. All our inferred physical properties of the system are listed in Table 4. We summarize our main results below.

- The optical counterpart to IGR J16194-2810 is high on the red giant branch (Figure 1). Given the giant's apparent magnitude of $G = 11.4$, the chance alignment of the optical counterpart in the field of IGR J16194-2810 with the X-ray source detected by Chandra is $\lesssim 3 \times 10^{-6}$. This, combined with the fact that we find the companion to be a dark object with mass $\approx 1.2\ M_\odot$ and the giant to be nearly Roche lobe-filling, confirms beyond reasonable doubt that the system is a SyXB.

- Based on ASAS-SN optical light curves displaying ellipsoidal modulation on a ≈ 96 day variability period (Figure 2), we infer that the red giant fills $\approx 82\%$ of its Roche lobe. We confirm a tentative detection of a spin period of 4.055 hours in the second half of Sector 12 TESS data (Figure 8), but do not consider this measurement to be definitive.

- By fitting an SED model to archival photometry of the red giant companion (Figure 3), we derive its stellar parameters to be $T_{\rm eff} = 3723^{+6}_{-7}$ K, $\log\left(g/{\rm cm\ s^{-2}}\right) = 0.958^{+0.004}_{-0.009}$, and $[{\rm Fe/H}] = -0.12^{+0.02}_{-0.01}$ (Table 4). These values agree with those derived from our comparison of the high-resolution optical MIKE spectrum to the GALAH spectral database to within 2σ (Figure 6). Based on MIST isochrones and single-star evolutionary models, we determine the red giant's mass, radius, and luminosity to be $0.99^{+0.02}_{-0.03}\ M_\odot$, 54.8 ± 0.7, $R_\odot$, and $518^{+13}_{-15}\ L_\odot$, respectively. We find that the observed ellipsoidal variability amplitude, when combined with the SED constraints, implies an orbital inclination of $(82^{+6}_{-7})^\circ$ (Figure 4).

- We perform follow-up with the FEROS and MIKE high-resolution spectrographs (Figure 6) and measure radial velocities over more than one orbital cycle (Figure 5). By fitting a Keplerian two-body orbit, we determine the orbital period of IGR J16194-2810 to be 192.73 ± 0.01 days. We derive a companion mass of $1.23^{+0.05}_{-0.03}\ M_\odot$ (Figure 7 and Table 4), dynamically confirming the presence of a neutron star in the symbiotic X-ray binary.

- Informed by the Galactic orbit of IGR J16194-2810 (Figure 9), we perform simulations to determine the distributions of allowed initial He star masses, orbital periods, and natal kicks that can reproduce the orbital parameters observed today (Figure 10). We also simulate the chance that the binary remains bound and at least as wide as observed following the supernova of the NS progenitor as a function of mass loss and natal kick (Figures 11 and 12), finding a non-zero probability of reproducing the current orbit for most reasonable initial configurations. We simulate the future evolution

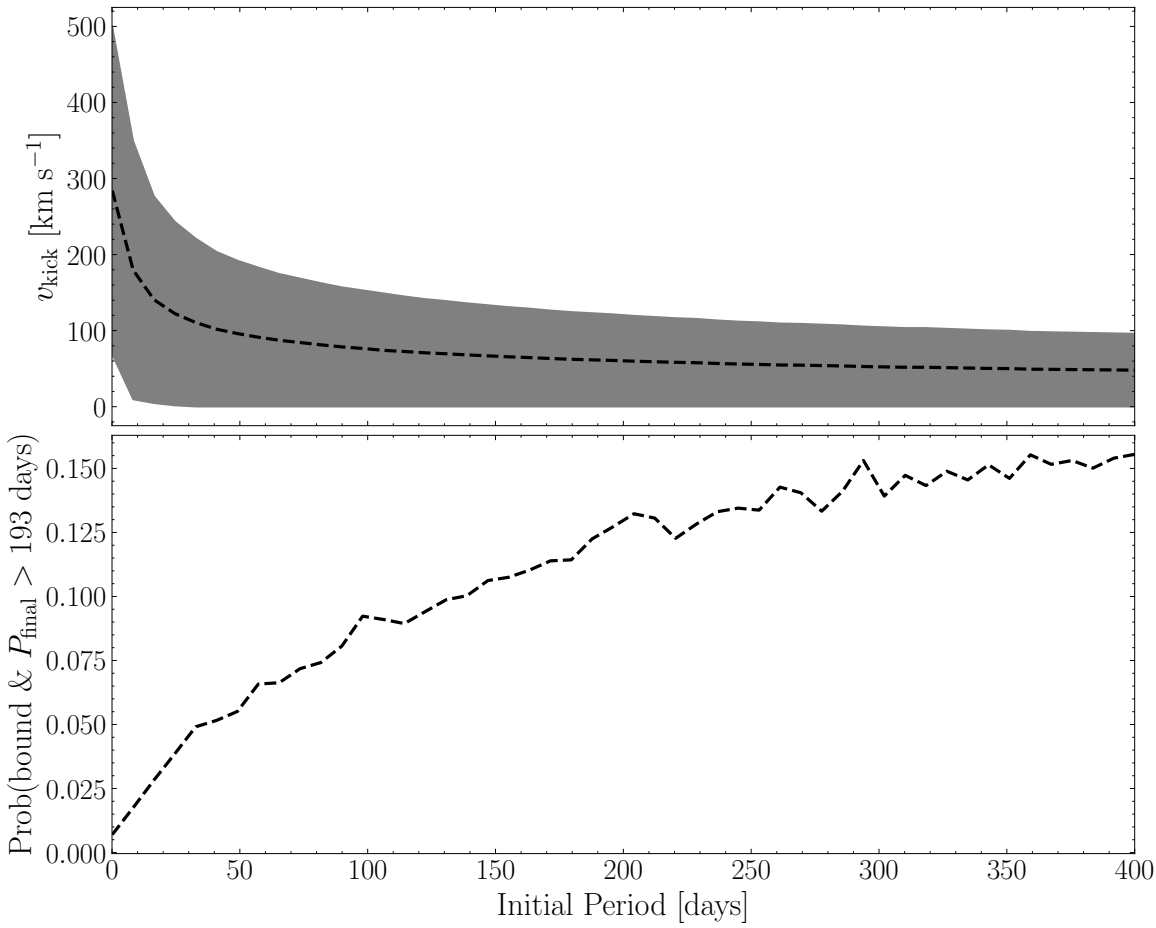

Figure 12. Top Panel: Range of natal kicks that allow the final orbit to both be bound and as least as wide as observed as a function of initial orbital period. Bottom Panel: Probability that orbit stays bound and is at least as wide as observed as a function of initial orbital period for median allowed natal kick. We assume a default He star mass of 3 $M_\odot$, a red giant mass of 1 $M_\odot$, and a neutron star mass of 1.4 $M_\odot$. At low initial orbital periods, a large natal kick is required to reach a final orbital separation at least as wide as observed, resulting in a low probability of survival for the resulting binary.

of the binary using MESA (Figure 13), concluding that the system will end up as a $\sim 0.4\,M_\odot$ helium WD orbiting the NS in a ~ 900 day orbit.

- Comparing IGR J16194-2810 (and other SyXBs) to the population of wide NS + MS binaries discovered in *Gaia* DR3, we conclude that SyXBs with $P_{\rm orb} \lesssim 1000$ d likely have a space density that is lower by a factor of $\gtrsim 100$ than wide NS + MS binaries. However, their X-ray bright lifetime is $\sim$ 1000 times shorter than the lifetime of typical NS+MS binaries, implying a comparable or higher birth rate. As a result, the population of SyXBs provides a window on the future evolution of these unexpectedly wide NS binaries.

7.1. *Comparison with Hinkle et al. (2024)*

In a contemporaneous analysis, Hinkle et al. (2024) utilize optical and near-infrared spectroscopy to derive a spectroscopic orbit for IGR J16194-2810. They also analyze the ellipsoidal variability of the red giant and infer its stellar parameters and abundances. While they do not jointly fit all available data as we do in Section 5.6, they derive orbital elements and stellar parameters in good agreement with our work. We compare our constraints against theirs in Table 5.

Hinkle et al. (2024) do not attempt to measure the NS mass. We constrain it to $M_{\rm NS} = 1.23^{+0.05}_{-0.03}\,M_\odot$. They assume the red giant exactly fills its Roche lobe when inferring its mass, but our analysis finds that it is only $\approx 82\%$ Roche lobe filling and that mass transfer is primarily via winds. The long spin period inferred from the TESS light curve ($P \approx 4.06$ hours; Luna 2023) supports this interpretation, since mass transfer by Roche lobe overflow is expected to rapidly spin up the NS.

In contrast to our SED fitting and interpolation on MIST stellar models, Hinkle et al. (2024) measure a metallicity directly from near-IR spectra. Their result of $[{\rm Fe/H}] = -0.14 \pm 0.12$ is in excellent agreement with our value of $[{\rm Fe/H}] = -0.12^{+0.02}_{-0.01}$.

Hinkle et al. (2024) also model the evolution of the system into a NS-WD binary. While they share our conclusions about the ultimate fate of IGR J16194-2810, they assume that the NS formed from a WD via accretion-induced collapse.

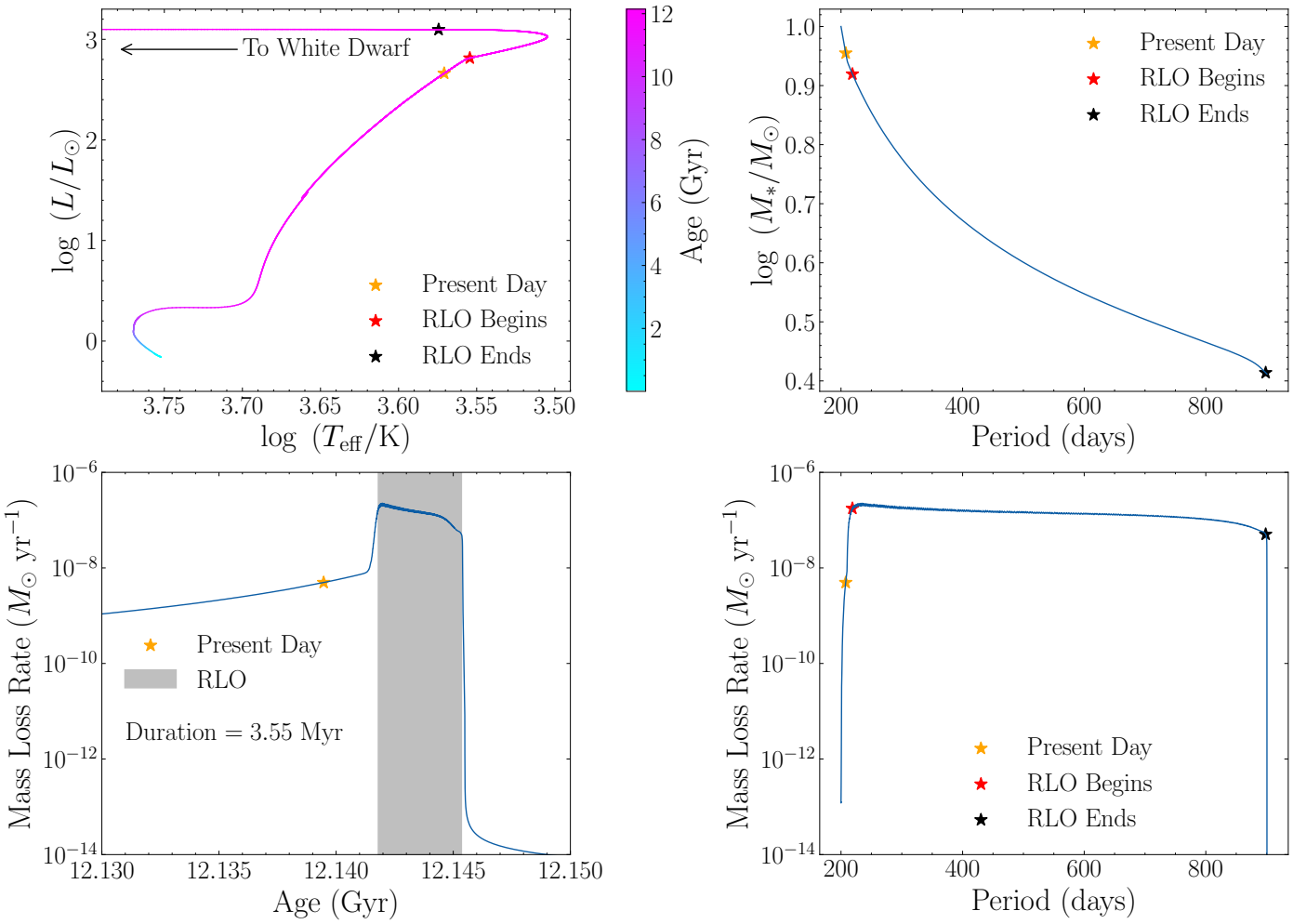

Figure 13. Top Left Panel: Evolution of donor star on a H-R diagram, with the luminosity and effective temperature at present day marked with an orange star. We focus on the main sequence and red giant branch. The slope of the red giant branch changes at the onset of Roche lobe overflow (marked with a red star). Top Right Panel: Mass of the red giant star as a function of orbital period. The star loses $\sim 0.6\ M_\odot$ over the course of its lifetime, the bulk of which is during Roche lobe overflow. This mass loss causes the orbit of the binary to widen significantly, eventually reaching ~ 900 days when the red giant loses its envelope and becomes a $0.4\ M_\odot$ white dwarf. Bottom Left Panel: Mass loss rate as a function of age of the red giant. The Roche lobe overflow phase (which lies about 5.88 Myr in the future of the binary and is shaded in gray) lasts for around 3.55 Myr. In the other panels, the start and end of Roche lobe overflow are marked with a red and black star, respectively. Bottom Right Panel: Mass loss rate as a function of orbital period. The elevated mass loss rate during Roche lobe overflow ($\sim 2 \times 10^{-7}\ M_\odot$ yr^{-1}) is responsible for the widening of the orbit from ~ 200 days to ~ 900 days.

We are agnostic about the formation process and show that system could have survived a NS natal kick of $\lesssim 50$ km s^{-1}. Furthermore, we provide additional context elucidating the connection between this SyXB and the population of wide NS-luminous star binaries discovered with *Gaia* astrometry.

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